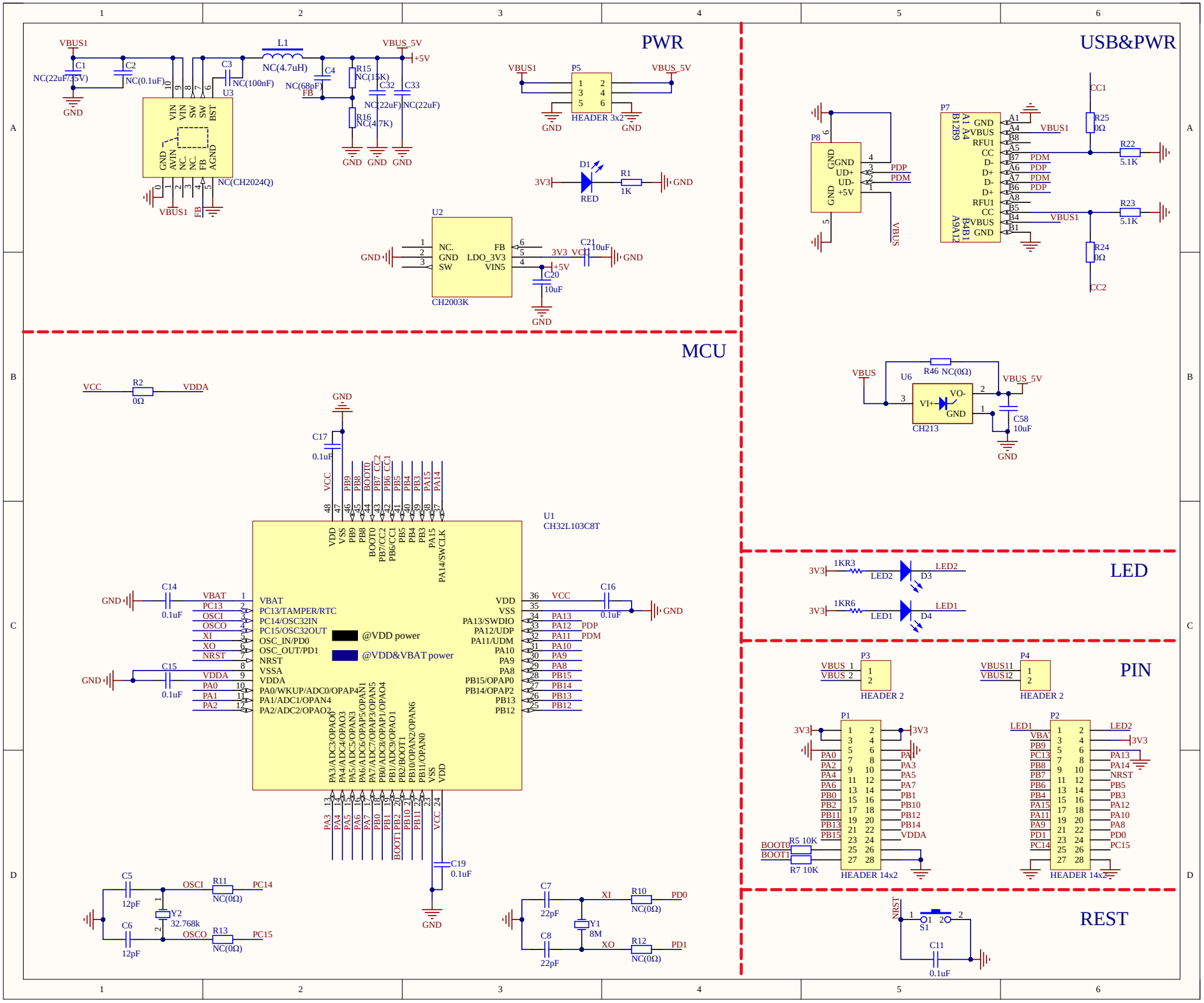
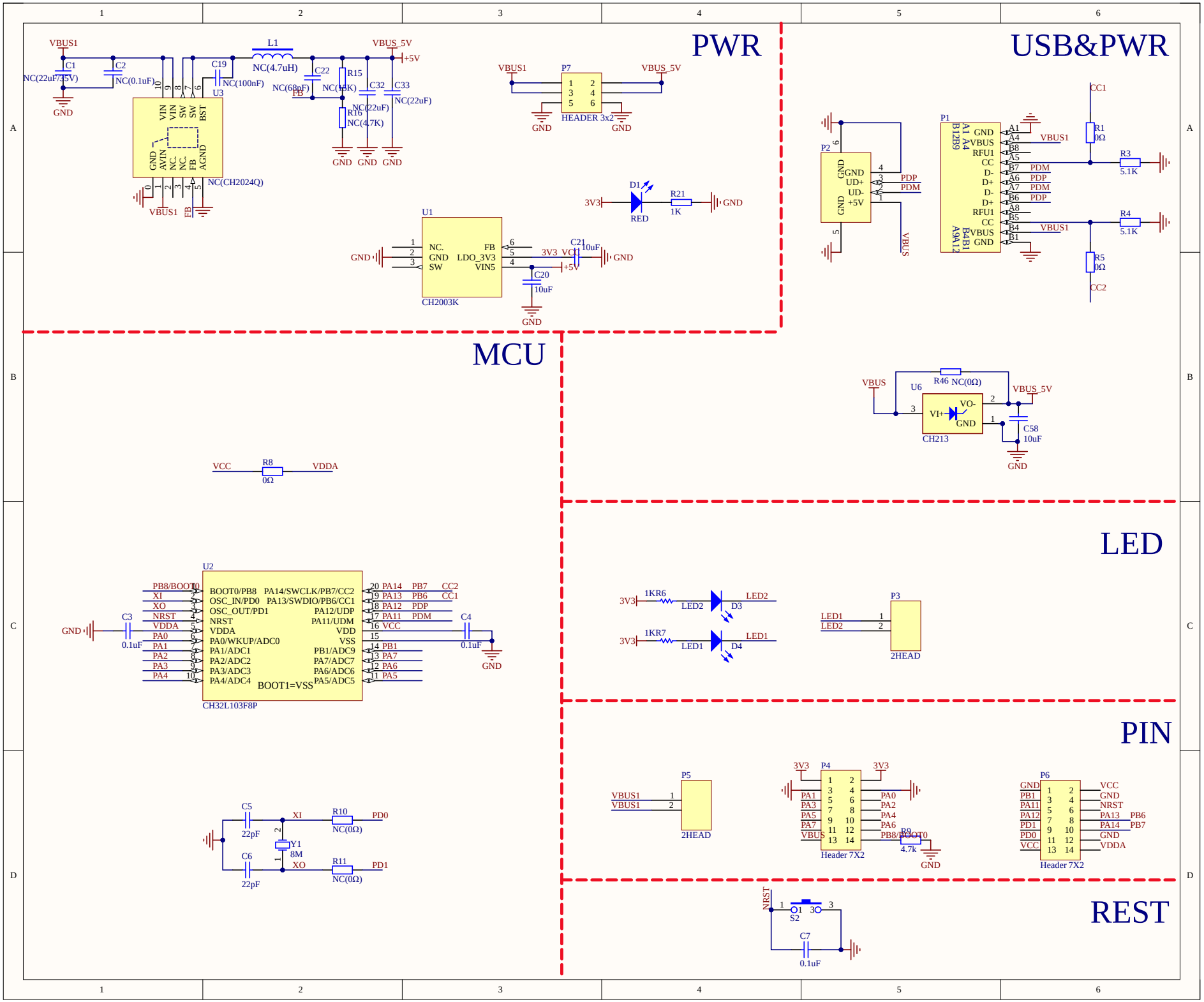
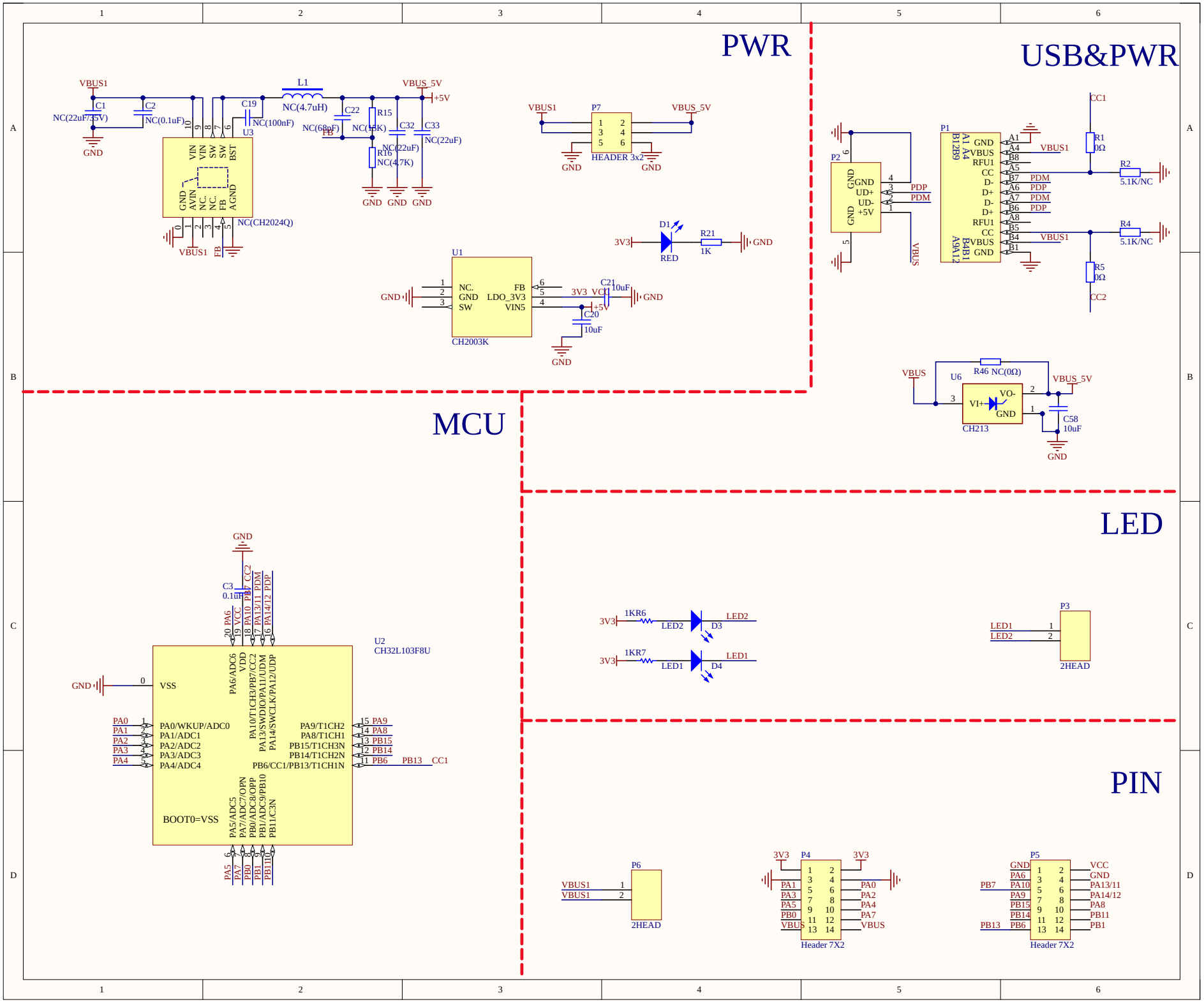
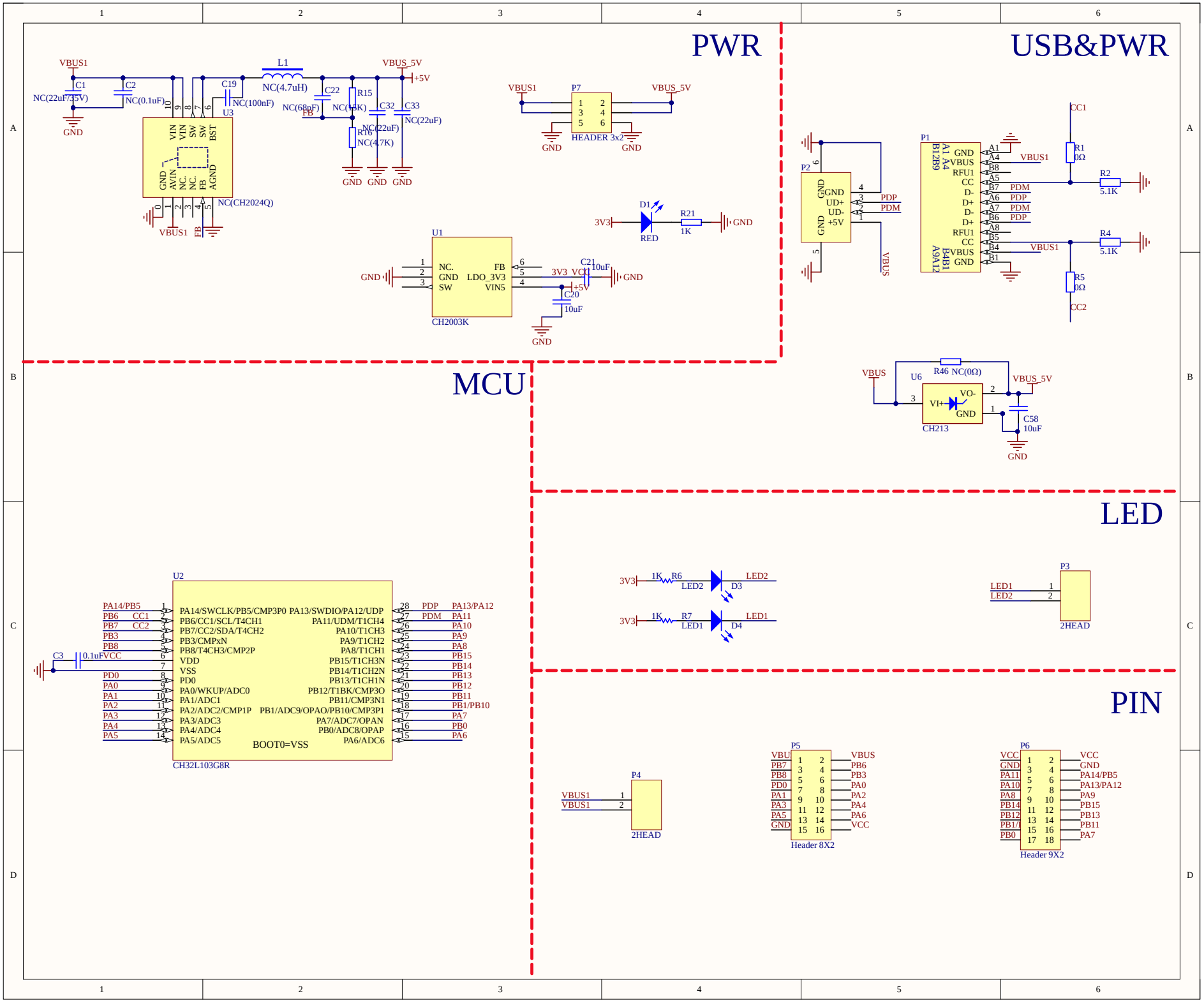
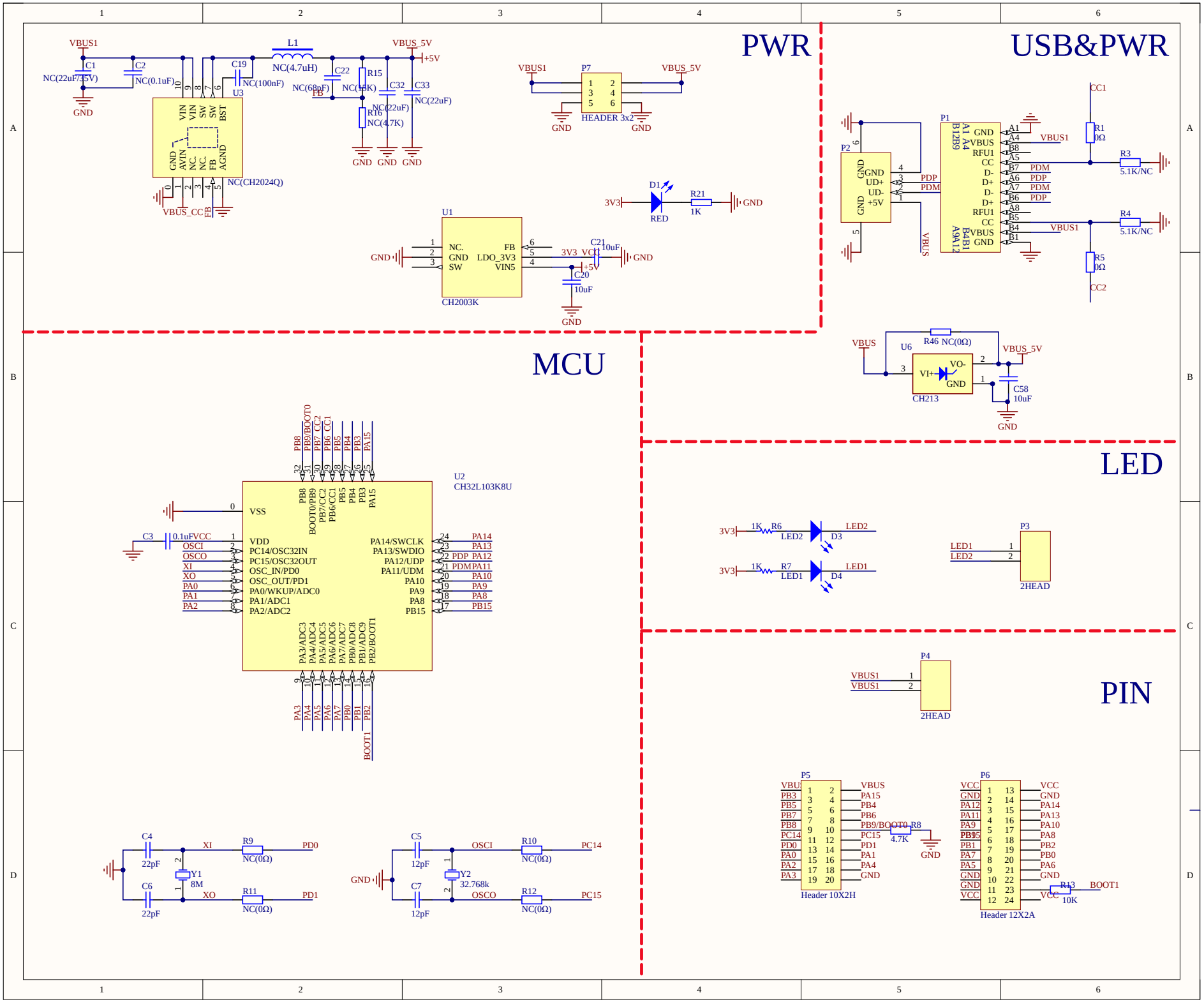

PWR
USB&PWR
MCU
LED
PIN
REST
VBUS1
C1
NC(22uF/35V)
C2
NC(0.1uF)
GND
L1
NC(4.7uH)
C3
NC(100nF)
U3
VIN
VIN
SW
SW
BST
GND
AVIN
NC.
NC.
FB
AGND
NC(CH2024Q)
C4
NC(68pF)
FB
R15
NC(15K)
R16
NC(47K)
C32
NC(22uF)
C33
NC(22uF)
VBUS_5V
+5V
P5
HEADER 3x2
D1
RED
R1
1K
3V3
U2
NC.
GND
SW
FB
LDO_3V3
VIN5
CH2003K
C21
10uF
C20
10uF
P8
GND
UD+
UD-
+5V
PDP
PDM
VBUS
P7
B12B9
A1 A4
B4B1
A9A12
GND
VBUS
RFU1
CC
D-
D+
D-
D+
RFU1
CC
VBUS
GND
PDM
PDP
PDM
PDP
CC1
CC2
R25
0Ω
R24
0Ω
R22
5.1K
R23
5.1K
VCC
R2
0Ω
VDDA
C17
0.1uF
U1
CH32L103C8T
VBAT
PC13/TAMPER/RTC
PC14/OSC32IN
PC15/OSC32OUT
OSC_IN/PD0
OSC_OUT/PD1
NRST
VSSA
VDDA
PA0/WKUP/ADC0/OPAP4N
PA1/ADC1/OPAN4
PA2/ADC2/OPAO2
@VDD power
@VDD&VBAT power
VDD
VSS
PA13/SWDIO
PA12/UDP
PA11/UDM
PA10
PA9
PA8
PB15/OPAP0
PB14/OPAP2
PB13
PB12
C14
C15
C16
0.1uF
C5
12pF
C6
12pF
Y2
32.768k
OSCI
OSCO
R11
NC(0Ω)
R13
NC(0Ω)
PC14
PC15
C7
22pF
C8
22pF
Y1
8M
XI
XO
R10
NC(0Ω)
R12
NC(0Ω)
PD0
PD1
C19
0.1uF
U6
CH213
R46 NC(0Ω)
C58
10uF
1KR3
1KR6
LED2
LED1
D3
D4
P3
P4
HEADER 2
P1
P2
HEADER 14x2
R5 10K
R7 10K
BOOT0
BOOT1
NRST
S1
C11
0.1uF


PWR
USB&PWR
MCU
LED
PIN
REST
VBUS1
VBUS_5V
+5V
L1
NC(4.7uH)
C1
NC(22uF/35V)
C2
NC(0.1uF)
C19
NC(100nF)
C22
NC(68nF)
R15
NC(15K)
C32
NC(22uF)
C33
NC(22uF)
R16
NC(4.7K)
U3
NC(CH2024Q)
GND
FB
P7
HEADER 3x2
U1
CH2003K
NC.
GND
SW
FB
LDO_3V3
VIN5
3V3_VCC
C21
10uF
C20
10uF
D1
RED
R21
1K
P2
P1
CC1
CC2
R1
0Ω
R3
5.1K
R4
5.1K
R5
0Ω
PDP
PDM
VBUS
U6
R46 NC(0Ω)
CH213
C58
10uF
VCC
R8
0Ω
VDDA
U2
CH32L103F8P
BOOT1=VSS
C3
0.1uF
C4
0.1uF
C5
22pF
C6
22pF
Y1
8M
XI
XO
R10
NC(0Ω)
R11
NC(0Ω)
PD0
PD1
1KR6
1KR7
LED2
LED1
D3
D4
P3
2HEAD
P5
P4
Header 7X2
P6
R9
4.7k
NRST
S2
C7
0.1uF


PWR
USB&PWR
MCU
LED
PIN
VBUS1
C1
NC(22uF/25V)
C2
NC(0.1uF)
GND
L1
NC(4.7uH)
C19
NC(100nF)
U3
NC(CH2024Q)
VIN VIN SW SW BST
GND AVIN NC. NC. FB AGND
VBUS_5V
+5V
C22
NC(68pF)
R15
NC(15K)
R16
NC(4.7K)
C32
NC(22uF)
C33
NC(22uF)
P7
HEADER 3x2
U1
CH2003K
NC. FB
GND LDO_3V3
SW VIN5
3V3
C21
10uF
C20
10uF
D1
RED
R21
1K
P2
P1
B12B9
A1 A4
B4B1
A9A12
PDP
PDM
CC1
CC2
R1
0Ω
R2
5.1K/NC
R4
5.1K/NC
R5
0Ω
U6
R46 NC(0Ω)
VO-
VI+
CH213
C58
10uF
U2
CH32L103F8U
VSS
PA0/WKUP/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA6/ADC6
VDD
PA10/T1CH3/PB7/CC2
PA13/SWDIO/PA11/UDM
PA14/SWCLK/PA12/UDP
PA9/T1CH2
PA8/T1CH1
PB15/T1CH3N
PB14/T1CH2N
PB6/CC1/PB13/T1CH1N
BOOT0=VSS
PA5/ADC5
PA7/ADC7/OPN
PB0/ADC8/OPP
PB1/ADC9/PB10
PB11/C3N
C3
0.1uF
1KR6
LED2
D3
1KR7
LED1
D4
P3
2HEAD
P6
P4
Header 7X2
P5


PWR
USB&PWR
MCU
LED
PIN
VBUS1
C1
NC(22uF/35V)
C2
NC(0.1uF)
GND
U3
VIN VIN SW SW BST
GND AVIN NC. NC. FB AGND
NC(CH2024Q)
L1
NC(4.7uH)
C19
NC(100nF)
VBUS_5V
+5V
C22
NC(68pF)
R15
NC(4K)
C32
NC(22uF)
C33
NC(22uF)
R16
NC(4.7K)
FB
P7
HEADER 3x2
U1
NC.
GND
SW
FB
LDO_3V3
VIN5
CH2003K
C21
10uF
C20
10uF
3V3
D1
RED
R21
1K
P2
GND
UD+
UD-
+5V
PDP
PDM
VBUS
P1
GND
VBUS
RFU1
CC
D-
D+
RFU1
CC
VBUS
GND
CC1
CC2
R1
0Ω
R2
5.1K
R4
5.1K
R5
0Ω
U6
CH213
R46 NC(0Ω)
VO-
VI+
GND
C58
10uF
U2
CH32L103G8R
PA14/PB5
PB6 CC1
PB7 CC2
PB3
PB8
VCC
VDD
VSS
PD0
PA0
PA1
PA2
PA3
PA4
PA5
PA14/SWCLK/PB5/CMP3P0
PB6/CC1/SCL/T4CH1
PB7/CC2/SDA/T4CH2
PB3/CMPxN
PB8/T4CH3/CMP2P
PD0
PA0/WKUP/ADC0
PA1/ADC1
PA2/ADC2/CMP1P
PA3/ADC3
PA4/ADC4
PA5/ADC5
BOOT0=VSS
PA13/SWDIO/PA12/UDP
PA11/UDM/T1CH4
PA10/T1CH3
PA9/T1CH2
PA8/T1CH1
PB15/T1CH3N
PB14/T1CH2N
PB13/T1CH1N
PB12/T1BK/CMP3O
PB11/CMP3N1
PB1/ADC9/OPAO/PB10/CMP3P1
PA7/ADC7/OPAN
PB0/ADC8/OPAP
PA6/ADC6
PDP PA13/PA12
PDM PA11
PA10
PA9
PA8
PB15
PB14
PB13
PB12
PB11
PB1/PB10
PA7
PB0
PA6
C3
0.1uF
3V3
1K
R6
LED2
D3
R7
LED1
D4
P3
2HEAD
P4
2HEAD
P5
Header 8X2
P6
Header 9X2


PWR
USB&PWR
MCU
LED
PIN
VBUS1
C1
NC(22uF/35V)
C2
NC(0.1uF)
GND
L1
NC(4.7uH)
C19
NC(100nF)
U3
VIN VIN SW SW BST
GND AVIN NC. NC. FB AGND
NC(CH2024Q)
VBUS_CC
FB
VBUS_5V
+5V
C22
NC(68pF)
R15
NC(15K)
R16
NC(4.7K)
C32
NC(22uF)
C33
NC(22uF)
P7
HEADER 3x2
U1
NC. GND SW
FB LDO_3V3 VIN5
CH2003K
3V3
C21
10uF
C20
10uF
D1
RED
R21
1K
P1
P2
GND UD+ UD- +5V
PDP
PDM
VBUS
CC1
CC2
R1
0Ω
R3
5.1K/NC
R4
5.1K/NC
R5
0Ω
U6
R46 NC(0Ω)
VI+ VO- GND
CH213
C58
10uF
U2
CH32L103K8U
VSS
VDD
OSCI
OSCO
XI
XO
PA0
PA1
PA2
BOOT0/PB9
PB7/CC2
PB6/CC1
PB5
PB4
PB3
PA15
PA14/SWCLK
PA13/SWDIO
PA12/UDP
PA11/UDM
PA10
PA9
PA8
PB15
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PB0/ADC8
PB1/ADC9
PB2/BOOT1
C3
0.1uF
VCC
C4
22pF
C6
22pF
Y1
8M
R9
NC(0Ω)
R11
NC(0Ω)
PD0
PD1
C5
12pF
C7
12pF
Y2
32.768k
R10
NC(0Ω)
R12
NC(0Ω)
PC14
PC15
R6
R7
LED2
LED1
D3
D4
P3
2HEAD
P4
2HEAD
P5
Header 10X2H
R8
4.7K
P6
Header 12X2A
R13
10K
BOOT1